

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 3– Visualization Design Activity
Weightage:	35%	Total Marks:	

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Section Batch: /	B.Tech AI and DS	Date of Submission:	20-08-2026
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I Dharshini H, 192324268 (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: Dharshini H

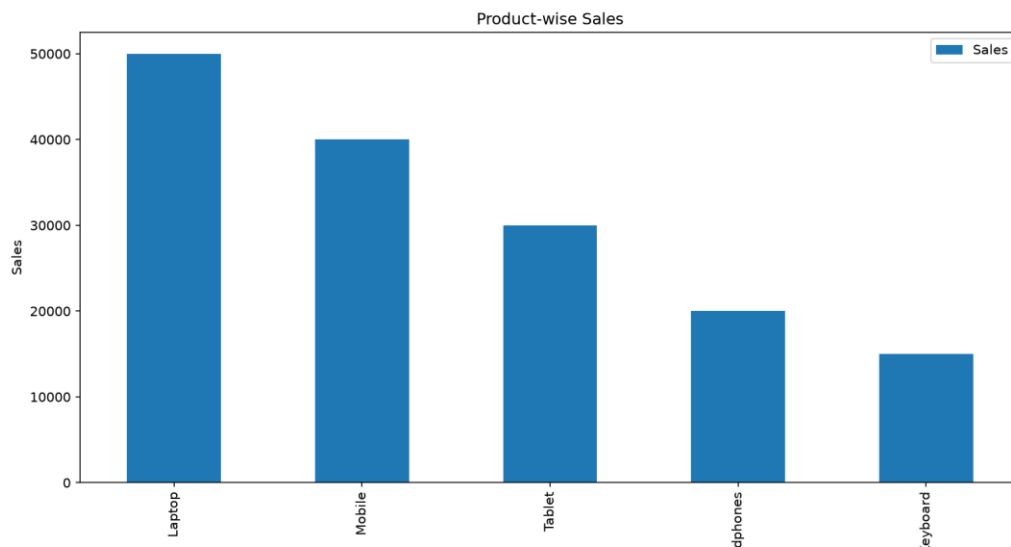
4	Pandas Visualization Activity	Create a DataFrame containing Product, Sales, and Profit. Generate appropriate visualizations directly using Pandas.
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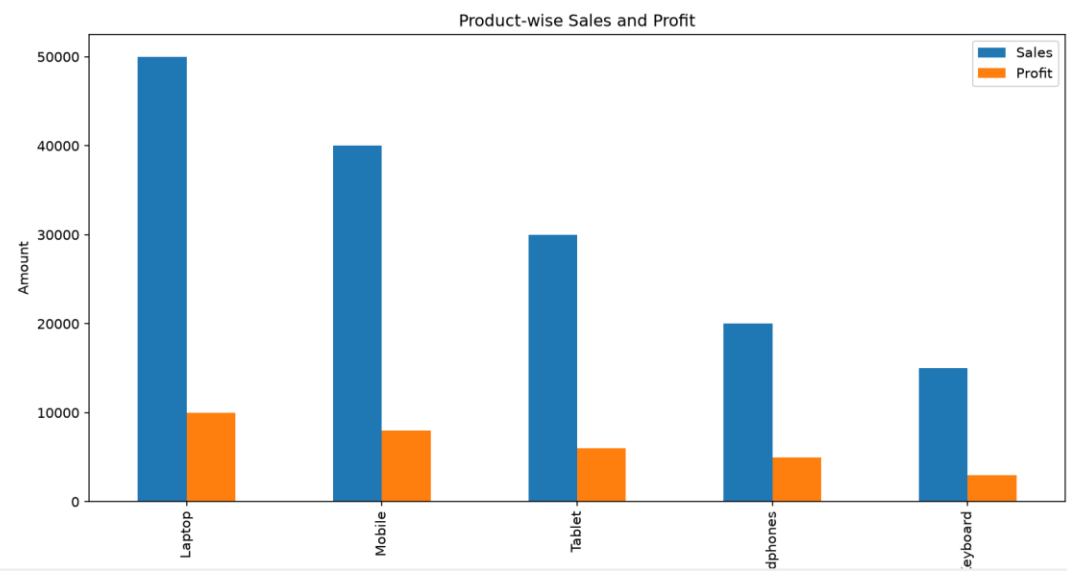
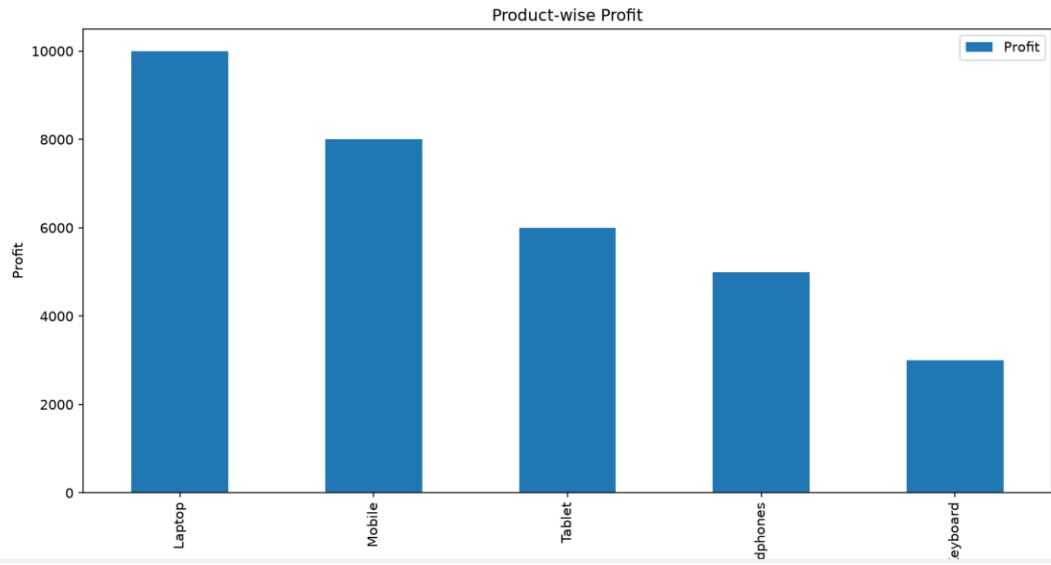
CODE:

```
COS > at3_1.py > ...
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 df = pd.DataFrame({
4     'Product': ['Laptop', 'Mobile', 'Tablet', 'Headphones', 'Keyboard'],
5     'Sales': [50000, 40000, 30000, 20000, 15000],
6     'Profit': [10000, 8000, 6000, 5000, 3000]
7 })
8 print(df)
9
10 # Plot Sales using Pandas
11 df.plot(x='Product', y='Sales', kind='bar', title='Product-wise Sales')
12 plt.show()
13
14 # Plot Profit using Pandas
15 df.plot(x='Product', y='Profit', kind='bar', title='Product-wise Profit')
16 plt.show()
17
18 # Plot Sales and Profit together
19 df.plot(x='Product', y=['Sales', 'Profit'], kind='bar',
20         title='Product-wise Sales and Profit')
21 plt.show()
```

OUTPUT:

	Product	Sales	Profit
0	Laptop	50000	10000
1	Mobile	40000	8000
2	Tablet	30000	6000
3	Headphones	20000	5000
4	Keyboard	15000	3000





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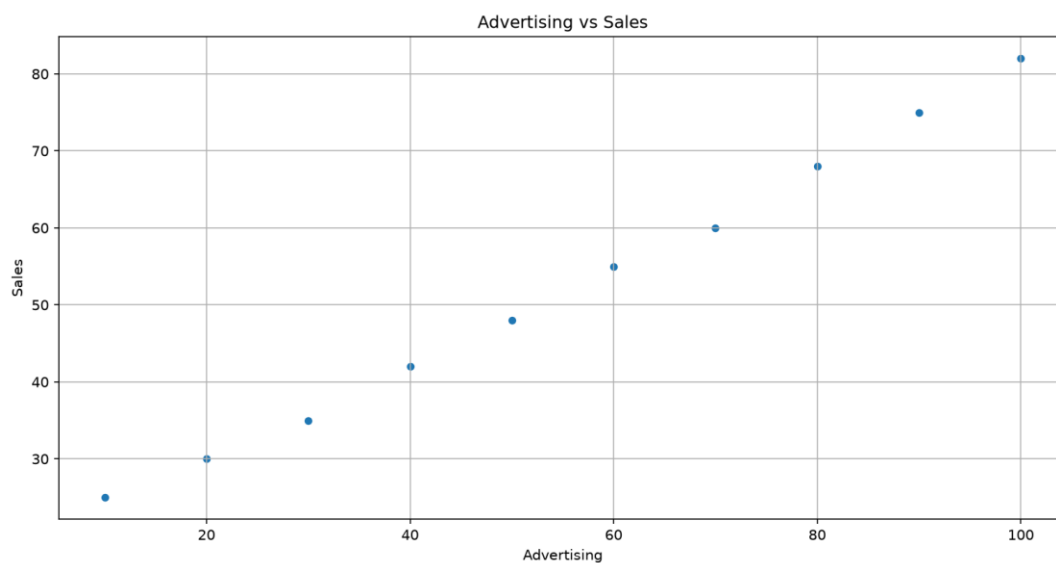
Relationship Visualization

Plot two variables such as **Advertising vs Sales** using a scatter plot. Determine whether the relationship appears positive, negative, or weak.

CODE:

```
COS > at3_2.py > ...
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 df = pd.DataFrame({
4     'Advertising': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100],
5     'Sales': [25, 30, 35, 42, 48, 55, 60, 68, 75, 82]
6 })
7 df.plot(
8     x='Advertising',
9     y='Sales',
10    kind='scatter',
11    title='Advertising vs Sales'
12 )
13 plt.grid(True)
14 plt.show()
15 # Calculate correlation
16 correlation = df['Advertising'].corr(df['Sales'])
17 print("Correlation:", correlation)
18 if correlation > 0.5:
19     print("Relationship: Positive")
20 elif correlation < -0.5:
21     print("Relationship: Negative")
22 else:
23     print("Relationship: Weak")
```

OUTPUT:



```
Correlation: 0.9985789846129846
Relationship: Positive
```